The effect of cation charge and concentration on the salt-induced deswelling of sodium polyacrylate

IB Chemistry Higher Level — Internal Assessment

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1. Introduction

Hydrogels are an area of innovation in biomedical chemistry because their cross-linked polymer chains retain large amounts of water, suggesting uses such as wound dressings. Whilst their interaction with pure water has been closely examined, a dressing is in practice exposed to electrolyte solutions such as wound exudate, which contain Na^+, K^+, Mg^{2+} and Ca^{2+}. A polymer that absorbs 120 times its mass of water may absorb far less of a body fluid.

This investigation therefore studies how the concentration of dissolved salt, and the charge and identity of its cation, change the swelling of sodium polyacrylate, the superabsorbent polymer used in nappies and some dressings. The anion (Cl^-) is kept constant so the cations can be compared directly.

2. Research question

How does increasing the concentration (0.00–0.20 mol dm^{-3}) of aqueous NaCl, KCl, $MgCl_2$ and $CaCl_2$ affect the apparent swelling degree of sodium polyacrylate, measured as its percentage mass gain after 120 s of immersion?

3. Background

3.1 Structure and swelling of sodium polyacrylate

Sodium polyacrylate is a lightly cross-linked network of poly(acrylic acid) chains in which many carboxylic acid groups are present as sodium carboxylate, $–[CH_2–CH(COO^-Na^+)]_n–$. In water, the Na^+ counter-ions dissociate from the fixed $–COO^-$ groups but cannot leave the network, because the solution outside must stay electrically neutral. Three effects then drive swelling:

1. **Osmotic pressure.** The mobile Na^+ ions trapped inside the gel create a much higher concentration of dissolved particles inside than outside, so water enters by osmosis (a Donnan equilibrium).
2. **Electrostatic repulsion.** Neighbouring $–COO^-$ groups repel each other and stretch the chains apart.
3. **Mixing.** Polar carboxylate groups are hydrated by water through ion–dipole interactions.

The cross-links oppose expansion with an elastic restoring force. Swelling continues until this force balances the swelling pressure.

3.2 Effect of salt concentration

Adding salt to the external solution raises its ion concentration and reduces the concentration *difference* across the gel boundary, which lowers the osmotic driving

force. Cations entering the gel also cluster around –COO⁻ groups and screen their charges, reducing repulsion. Both effects depend on the ionic strength I of the solution:

$$I = 1/2 \sum_i c_i z_i^2$$

where c_i is the concentration and z_i the charge of each ion. For NaCl, $I = c$, but for $CaCl_2$, $I = ½(4c + 2c) = 3c$.

Flory (1953) combined these ideas into an expression for the equilibrium volume swelling ratio Q of an ionic gel. When the ionic term dominates, it reduces to

$$Q^{5/3} \propto \frac{1}{I} \Rightarrow Q \propto I^{-3/5}$$

For a highly swollen gel the percentage mass gain S is approximately proportional to Q, and for a single salt $I \propto c$, so

$$S = k\,c^n \Rightarrow \ln S = n \ln c + \ln k\,, n_{\text{Flory}} = -0.60$$

A graph of ln S against ln c should therefore be linear with gradient n. This limiting law assumes equilibrium, and an external salt concentration much larger than the fixed-charge concentration inside the gel; both are examined in Section 6.

3.3 Effect of cation charge and identity

Divalent cations screen more strongly: at equal concentration they give a larger ionic strength ($3c$ compared with c). They can also bind *two* –COO⁻ groups on different chains at once, forming ionic cross-links that tighten the network. This specific binding is not described by ionic strength alone.

Cation identity within each charge group also matters. Table 1 shows that Mg^{2+} has a higher charge density than Ca^{2+} and a more exothermic hydration enthalpy, so it holds its hydration shell more tightly. Vazquez de Vasquez et al. (2021) found that Ca^{2+} forms direct contact ion pairs with carboxylate, whereas Mg^{2+} rarely does and remains separated from –COO⁻ by water molecules. Similarly, Vrbka et al. (2006) showed that Na^+ pairs with carboxylate more strongly than K^+ does.

Table 1. Properties of the four cations (IBO, 2023).

Ion	Charge	Ionic radius / pm	Charge ÷ radius / pm⁻¹	Enthalpy of hydration / kJ mol⁻¹
Na^+	+1	102	0.0098	−424
K^+	+1	138	0.0072	−340
Mg^{2+}	+2	72	0.0278	−1963
Ca^{2+}	+2	100	0.0200	−1616

3.4 Hypotheses

1. Swelling will decrease as salt concentration increases for all four salts, with ln S against ln c approximately linear and gradient near −0.60 for NaCl and KCl.
2. The divalent salts will reduce swelling much more than the monovalent salts, even when compared at *equal ionic strength*, because of ionic cross-linking.
3. $CaCl_2$ will reduce swelling more than $MgCl_2$, and NaCl slightly more than KCl, because of stronger direct cation–carboxylate pairing.

4. Methodology

4.1 Variables

Table 2. Variables.

Type	Variable	How it was controlled or measured	Why it matters
Independent	Salt concentration: 0.00, 0.05, 0.10, 0.15, 0.20 mol dm^{-3}	Salt weighed (±0.001 g) and made up to 50.00 cm^3 in a volumetric flask; actual concentration calculated from actual mass	Five levels allow a trend and a power-law fit; the range includes the ionic strength of body fluids (≈0.15 mol dm^{-3})
Independent	Cation: Na^+, K^+, Mg^{2+}, Ca^{2+}	Chloride salt of each cation used	Compares charge (+1 against +2) and, within each charge, ion size and hydration
Dependent	Apparent swelling degree, S / %	S = (F − T)/(T − B) × 100 from three balance readings	Mass of solution absorbed per unit mass of dry polymer
Controlled	Anion (Cl^-)	All salts are chlorides	Anions also affect swelling; a common anion isolates the cation effect
Controlled	Dry polymer mass (0.050 ± 0.002 g)	Weighed into the bag; S uses each trial's actual mass	Sample size changes how tightly the swollen gel fills the bag
Controlled	Solution volume (50.00 cm^3)	Whole contents of one volumetric flask	Keeps cations in large excess so the external concentration stays constant
Controlled	Immersion time (120 s)	Stopwatch started as the bag was submerged	Swelling increases with time until equilibrium
Controlled	Draining and blotting	Drained until dripping stopped; three 30 s blots on fresh tissue	Surface liquid left on the bag would increase F
Controlled	Tea bag type and folding	Identical bags, each folded twice	The bag itself holds some solution; identical bags keep this similar
Controlled	Polymer	Same sodium polyacrylate product throughout	Cross-link density sets the maximum swelling

Type	Variable	How it was controlled or measured	Why it matters
Controlled (limited)	Temperature	All trials in the same laboratory; not measured	Swelling depends on temperature (see Section 8)

4.2 Apparatus and materials

Table 3. Apparatus and materials.

Item	Quantity	Uncertainty
Electronic balance	1	±0.001 g
Volumetric flask, 50.00 cm^3 (Class A)	1	±0.06 cm^3
Beakers, 50 cm^3 and 100 cm^3	1 each	–
Stopwatch	1	±0.01 s
Weighing boats, spatula, glass rod, funnel, wash bottle, tweezers	–	–
Empty paper tea bags	60	–
Paper tissue	–	–
Sodium polyacrylate granules	≈3 g	–
NaCl, KCl, anhydrous $CaCl_2$, anhydrous $MgCl_2$	≈3–6 g each	–
Distilled water	≈3 dm^3	–

4.3 Procedure

Solution preparation (a fresh solution for every trial)

1. Calculate the salt mass needed, $m = cVM$ with $V = 0.0500$ dm^3 (e.g. 0.146 g NaCl for 0.05 mol dm^{-3}).
2. Tare a weighing boat and add salt until the reading is within 0.005 g of the target. Record the actual mass.
3. Dissolve the salt in some distilled water in a 50 cm^3 beaker, stirring with a glass rod.
4. Transfer through a funnel into the 50.00 cm^3 volumetric flask, rinsing the beaker, rod and funnel into the flask.
5. Make up to the mark, stopper and invert several times. For 0.00 mol dm^{-3}, use distilled water only.

Sample preparation

6. Weigh an empty tea bag (B).
7. Add sodium polyacrylate with a spatula until the combined mass T exceeds B by 0.050 ± 0.002 g. Record T.

8. Shake the granules to the bottom of the bag and fold the top twice to close it.

Swelling measurement

9. Pour the whole solution into a 100 cm^3 beaker.
10. Submerge the bag fully with tweezers, starting the stopwatch at the same moment.
11. At 120 s, remove the bag and hold it against the beaker wall until dripping stops.
12. Place the bag on dry tissue for 30 s, then turn it onto a fresh dry area; repeat until it has been turned three times.
13. Weigh the bag immediately (F).
14. Repeat to give three trials at each concentration for each salt (60 trials).

4.4 Methodological considerations

- **Fresh solution per trial.** A preparation error affects only one trial.
- **Constant ion excess.** 50.00 cm^3 of 0.05 mol dm^{-3} solution contains 2.5×10^{-3} mol of cations, against at most 5.3×10^{-4} mol of $–COO^-$ in 0.050 g of polymer ($M = 94.04$ g mol^{-1} per repeating unit). With this four-fold excess, and under 5% of the solution absorbed, the external concentration stays approximately constant.
- **Sample size.** 0.050 g was chosen so that the gel swollen in water (about 6 g) could still expand inside the bag. Swelling was calculated from each trial's measured $T - B$, not the nominal 0.050 g, so small differences in sample size do not matter.
- **Three trials per condition** give a standard deviation for each mean within a manageable 60 trials.
- **Fixed immersion time.** 120 s allowed 60 trials to be completed consistently, so the result is a fixed-time (apparent) swelling, not an equilibrium value, as the research question specifies.

4.5 Safety, ethical and environmental considerations

Table 4. Safety and environmental considerations.

Hazard	Risk	Precaution / disposal
Sodium polyacrylate powder	Dust irritates eyes and airways; wet spills are very slippery	Safety glasses; transferred gently with a spatula; spills swept up while dry
Anhydrous $CaCl_2$	Serious eye irritant; dissolves strongly exothermically	Safety glasses; added to water gradually with stirring
NaCl, KCl, $MgCl_2$	Low hazard; mild eye irritants	Safety glasses; hands washed after use
Glassware	Cuts from breakage	Glassware checked for chips; breakages reported
Swollen gel (environmental)	Can block drains	Disposed of in solid waste, never the sink
Salt solutions (environmental)	Dilute (≤ 0.20 mol dm^{-3}) and of low toxicity	Poured to drain with plenty of water; salt masses kept small

No living organisms were involved, so there were no ethical concerns.

5. Results and data processing

5.1 Raw data

All masses were read from the same balance (±0.001 g). The four 0.00 mol dm^{-3} groups are distilled-water controls run within each series.

Table 5. Raw data for NaCl.

Target c / mol dm^{-3}	Trial	Salt mass, m / g ±0.001	Empty bag, B / g ±0.001	Bag + dry polymer, T / g ±0.001	Final mass, F / g ±0.001
0.00	1	0.000	0.468	0.519	6.786
0.00	2	0.000	0.483	0.532	6.583
0.00	3	0.000	0.477	0.528	7.010
0.05	1	0.143	0.478	0.527	2.634
0.05	2	0.147	0.485	0.534	2.588
0.05	3	0.147	0.490	0.541	2.731
0.10	1	0.292	0.488	0.537	2.045
0.10	2	0.294	0.482	0.531	2.098
0.10	3	0.290	0.483	0.534	2.117
0.15	1	0.440	0.479	0.530	1.896
0.15	2	0.439	0.479	0.528	1.865
0.15	3	0.439	0.489	0.540	1.879
0.20	1	0.586	0.488	0.538	1.710
0.20	2	0.582	0.484	0.535	1.676
0.20	3	0.584	0.484	0.533	1.570

Table 6. Raw data for KCl.

Target c / mol dm⁻³	Trial	Salt mass, m / g ±0.001	Empty bag, B / g ±0.001	Bag + dry polymer, T / g ±0.001	Final mass, F / g ±0.001
0.00	1	0.000	0.466	0.516	6.544
0.00	2	0.000	0.455	0.504	6.441
0.00	3	0.000	0.486	0.535	6.528
0.05	1	0.186	0.491	0.541	2.599
0.05	2	0.186	0.492	0.543	2.594
0.05	3	0.188	0.493	0.544	2.663
0.10	1	0.375	0.495	0.544	2.190
0.10	2	0.374	0.498	0.547	2.201
0.10	3	0.375	0.489	0.539	2.242
0.15	1	0.560	0.508	0.559	2.004
0.15	2	0.559	0.497	0.547	2.016
0.15	3	0.559	0.476	0.526	2.008
0.20	1	0.746	0.448	0.499	1.728
0.20	2	0.745	0.472	0.522	1.624
0.20	3	0.744	0.487	0.536	1.663

Table 7. Raw data for $CaCl_2$.

Target c / mol dm⁻³	Trial	Salt mass, m / g ±0.001	Empty bag, B / g ±0.001	Bag + dry polymer, T / g ±0.001	Final mass, F / g ±0.001
0.00	1	0.000	0.483	0.534	6.638
0.00	2	0.000	0.468	0.519	6.779
0.00	3	0.000	0.477	0.526	6.584
0.05	1	0.278	0.479	0.528	1.489
0.05	2	0.277	0.470	0.519	1.296
0.05	3	0.278	0.483	0.533	1.365
0.10	1	0.555	0.483	0.533	0.978
0.10	2	0.557	0.465	0.516	0.993
0.10	3	0.555	0.472	0.523	0.987
0.15	1	0.832	0.458	0.507	0.802
0.15	2	0.832	0.477	0.528	0.851
0.15	3	0.833	0.478	0.529	0.843
0.20	1	1.110	0.468	0.519	0.756
0.20	2	1.109	0.478	0.529	0.741
0.20	3	1.114	0.491	0.542	0.733

Table 8. Raw data for MgCl₂.

Target c / mol dm⁻³	Trial	Salt mass, m / g ±0.001	Empty bag, B / g ±0.001	Bag + dry polymer, T / g ±0.001	Final mass, F / g ±0.001
0.00	1	0.000	0.478	0.527	6.453
0.00	2	0.000	0.479	0.529	6.301
0.00	3	0.000	0.483	0.532	6.356
0.05	1	0.240	0.477	0.527	1.545
0.05	2	0.239	0.476	0.526	1.503
0.05	3	0.239	0.485	0.535	1.549
0.10	1	0.475	0.485	0.536	1.005
0.10	2	0.475	0.477	0.526	1.039
0.10	3	0.479	0.477	0.527	0.994
0.15	1	0.714	0.477	0.526	0.861
0.15	2	0.713	0.489	0.540	0.901
0.15	3	0.713	0.478	0.529	0.897
0.20	1	0.954	0.470	0.521	0.767
0.20	2	0.953	0.455	0.504	0.793
0.20	3	0.949	0.475	0.525	0.788

5.2 Processing

Concentration. The actual concentration was calculated from each measured salt mass. For NaCl, 0.05 mol dm⁻³, trial 1:

$$c = \frac{m}{M\,V} = \frac{0.143\,\text{g}}{58.44\,\text{g mol}^{-1} \times 0.0500\,\text{dm}^{3}} = 0.04894\,\text{mol dm}^{-3}$$

Swelling degree. With dry polymer mass $H = T - B$ and mass gain $D = F - T$, the apparent swelling degree is

$$S = \frac{D}{H} \times 100 = \frac{F - T}{T - B} \times 100$$

For the same trial: H = 0.527 − 0.478 = 0.049 g, D = 2.634 − 0.527 = 2.107 g, so

$$S = \frac{2.107}{0.049} \times 100 = 4300\,\%$$

That is, the polymer gained 43.0 times its own mass. S was calculated for every trial before averaging, and unrounded values were carried through.

Ionic strength. For CaCl₂ at a mean concentration of 0.05004 mol dm⁻³:

$$I = 1/2\left(0.05004 \times 2^{2} + 2 \times 0.05004 \times 1^{2}\right) = 0.1501\,\text{mol dm}^{-3}$$

5.3 Uncertainties

Propagated (apparatus) uncertainty. Each balance reading is ±0.001 g, so each difference has ±0.002 g. Relative uncertainties add for division:

$$\frac{\Delta S}{S}=\frac{\Delta H}{H}+\frac{\Delta D}{D}=\frac{0.002}{0.049}+\frac{0.002}{2.107}=0.0408+0.0009=0.0418$$

$$S=4300\pm180\,\%\,(4.2\,\%)$$

The dry polymer mass contributes 98% of this, because 0.002 g is 4% of a 0.050 g sample. Across all 60 trials the propagated uncertainty ranged from 4.0% to 5.0%.

For concentration, the relative uncertainties of the mass and the flask volume add:

$$\frac{\Delta c}{c}=\frac{0.001}{0.143}+\frac{0.06}{50.00}=0.0070+0.0012=0.0082 \Rightarrow c=0.04894\pm0.00040\,\mathrm{mol\ dm^{-3}}$$

Concentration and ionic-strength uncertainties are therefore below 1%, too small to show as horizontal error bars.

Random uncertainty. The sample standard deviation s describes the spread of the three trials. For NaCl at 0.05 mol dm⁻³:

$$\acute{S}=\frac{4300.0+4191.8+4294.1}{3}=4262.0\,\%, s=\sqrt{\frac{\sum(S_i-\acute{S})^2}{n-1}}=60.8\,\%$$

Uncertainty used. Each mean is quoted with the *larger* of s and the mean propagated uncertainty, so the uncertainty is not underestimated; here, 4260 ± 180% (4.1%, rounded to 2 significant figures). The propagated value was the larger in 14 of 20 conditions; the spread between trials was larger in the other six.

5.4 Processed results

Table 9. Processed results. The result's uncertainty is the larger of s and the propagated ΔS, rounded to 2 s.f.

Salt	Actual c / mol dm⁻³	S_1 / %	S_2 / %	S_3 / %	Mean / %	s / %	Propagated ΔS / %	Result / %
NaCl	0 (water)	12,288	12,349	12,710	12,449	228	499	12,450 ± 500
NaCl	0.0499 ± 0.0004	4,300	4,192	4,294	4,262	61	176	4,260 ± 180
NaCl	0.0999 ± 0.0005	3,078	3,198	3,104	3,126	63	130	3,130 ± 130
NaCl	0.1504 ± 0.0005	2,678	2,729	2,625	2,677	52	110	2,680 ± 110
NaCl	0.1999 ± 0.0006	2,344	2,237	2,116	2,233	114	93	2,230 ± 110
KCl	0 (water)	12,056	12,116	12,231	12,134	89	496	12,130 ± 500
KCl	0.0501 ± 0.0003	4,116	4,022	4,155	4,097	69	166	4,100 ± 170
KCl	0.1005 ± 0.0004	3,359	3,376	3,406	3,380	24	141	3,380 ± 140
KCl	0.1501 ± 0.0004	2,833	2,938	2,964	2,912	69	120	2,910 ± 120
KCl	0.1999 ± 0.0005	2,410	2,204	2,300	2,305	103	96	2,300 ± 100
$CaCl_2$	0 (water)	11,969	12,275	12,363	12,202	207	489	12,200 ± 490

Salt	Actual c / mol dm⁻³	S₁ / %	S₂ / %	S₃ / %	Mean / %	s / %	Propagated ΔS / %	Result / %
CaCl₂	0.0500 ± 0.0002	1,961	1,586	1,664	1,737	198	74	1,740 ± 200
CaCl₂	0.1001 ± 0.0003	890	935	910	912	23	40	912 ± 40
CaCl₂	0.1500 ± 0.0004	602	633	616	617	16	28	617 ± 28
CaCl₂	0.2002 ± 0.0004	465	416	375	418	45	20	418 ± 45
MgCl₂	0 (water)	12,094	11,544	11,886	11,841	278	484	11,840 ± 480
MgCl₂	0.0503 ± 0.0003	2,036	1,954	2,028	2,006	45	84	2,006 ± 84
MgCl₂	0.1001 ± 0.0003	920	1,047	934	967	70	43	967 ± 70
MgCl₂	0.1498 ± 0.0004	684	708	722	704	19	32	704 ± 32
MgCl₂	0.2000 ± 0.0005	482	590	526	533	54	25	533 ± 54

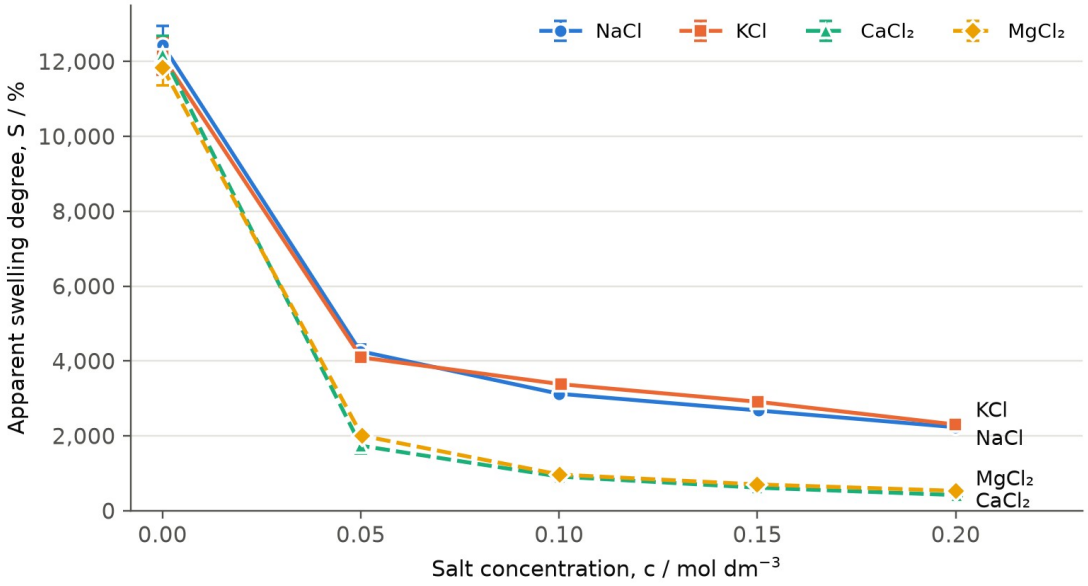


Figure 1. Mean apparent swelling degree against salt concentration. Error bars show the uncertainty in Table 9; some are smaller than the markers.

6. Analysis

6.1 Effect of concentration

Figure 1 shows that swelling fell with every increase in concentration for all four salts. The fall was not linear: most of it happened between 0.00 and 0.05 mol dm⁻³, where swelling dropped by 66% for NaCl and KCl and by 83–86% for MgCl₂ and CaCl₂. By 0.20 mol dm⁻³ the total reductions were 82% (NaCl), 81% (KCl), 96% (MgCl₂) and 97% (CaCl₂). This steep-then-shallow shape is consistent with the inverse power law in Section 3.2.

The twelve distilled-water controls gave a mean of 12,157% with a relative standard deviation of 2.4%, so the method was reproducible across the four series.

6.2 Power-law analysis

To test the Flory model, ln S was plotted against ln c for the non-zero concentrations (Figure 2). The uncertainty in ln S equals the relative uncertainty in S. The gradient uncertainty was found from the steepest and shallowest lines that pass through the error bars of the first and last points:

$$\Delta n = \frac{n_{\text{shallowest}} - n_{\text{steepest}}}{2} = \frac{-0.399 - (-0.532)}{2} = 0.066\,(\text{NaCl})$$

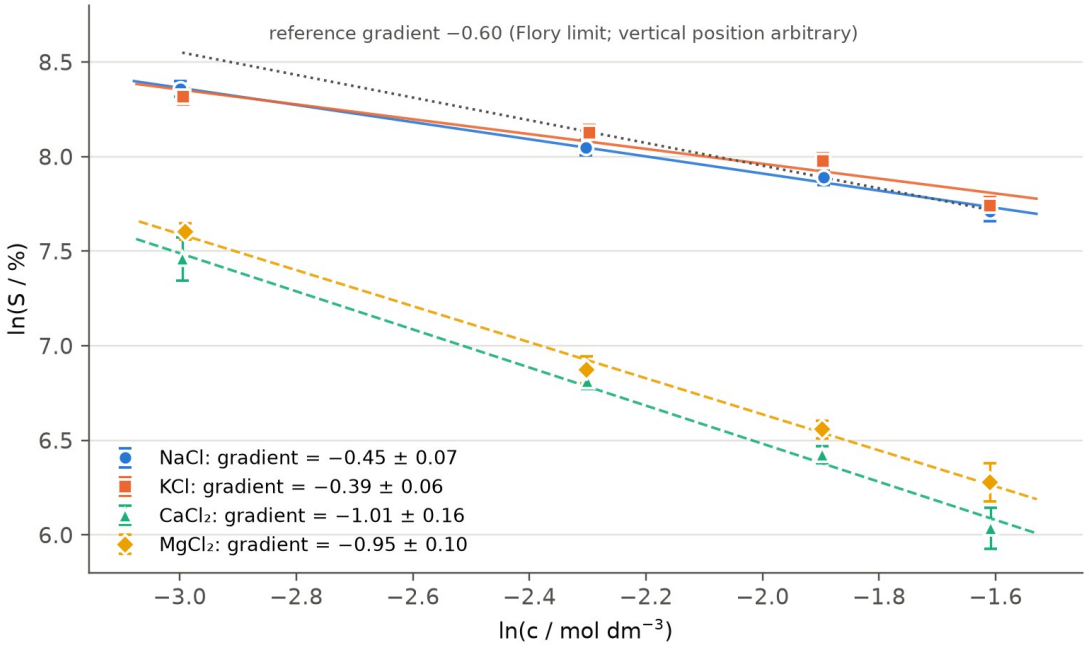


Figure 2. ln S against ln c with least-squares lines. The dotted line shows the Flory limiting gradient of −0.60 for comparison.

Table 10. Gradients of ln S against ln c.

Salt	Least-squares gradient	Steepest	Shallowest	n ± Δn	R²
NaCl	-0.453	-0.532	-0.399	-0.45 ± 0.07	0.994
KCl	-0.392	-0.477	-0.354	-0.39 ± 0.06	0.939
$CaCl_2$	-1.007	-1.187	-0.867	-1.01 ± 0.16	0.994
$MgCl_2$	-0.953	-1.064	-0.856	-0.95 ± 0.10	0.996
Flory limit	−0.60			−0.60	

All four plots are strongly linear ($R^2 \geq 0.94$), so swelling does follow a power law in concentration. However, the gradients split into two distinct groups:

- **Monovalent salts** (n ≈ −0.4 to −0.45). The gradients are 25% (NaCl) and 35% (KCl) *shallower* than −0.60, and neither uncertainty range (steepest −0.53) includes the Flory value. The limiting-law assumption explains part of

this. Taking all 5.3×10^{-4} mol of $–COO^-$ in the absorbed solution, the fixed-charge concentration inside the gel was about 0.25 mol dm^{-3} at 0.05 mol dm^{-3} NaCl and 0.48 mol dm^{-3} at 0.20 mol dm^{-3}. Even allowing for incomplete neutralisation of commercial polyacrylate, this is comparable to or larger than the external salt concentration, not much smaller as the model requires. In this regime swelling is less sensitive to added salt, so the gradient is shallower. The systematic errors in Section 8 act in the same direction.

- **Divalent salts** ($n \approx -1.0$). The gradients are about twice as steep as the monovalent ones and steeper than the Flory value. Screening cannot explain this, because Flory's model already includes it. The extra decrease supports ionic cross-linking: each Ca^{2+} or Mg^{2+} that bridges two $–COO^-$ groups both removes charge and tightens the network, and more bridges form as concentration rises.

6.3 Separating charge from ionic strength

At equal molarity, a divalent salt has three times the ionic strength of a monovalent salt, so Figure 1 does not isolate the effect of charge. Figure 3 therefore plots swelling against ionic strength. At $I = 0.15$ mol dm^{-3}, measured directly for all four salts (0.15 mol dm^{-3} NaCl and KCl; 0.05 mol dm^{-3} $CaCl_2$ and $MgCl_2$):

- $CaCl_2$ gave 1740 ± 200%, 35% lower than NaCl (2680 ± 110%).
- $MgCl_2$ gave 2006 ± 84%, 31% lower than KCl (2910 ± 120%).

The error bars do not overlap, so the divalent salts suppress swelling more than their ionic strength predicts, supporting hypothesis 2. Comparing solutions with the same *total* ion concentration (0.30 mol dm^{-3}) gives a larger difference: 912% for $CaCl_2$ (0.10 mol dm^{-3}) against 2680% for NaCl (0.15 mol dm^{-3}), a 66% reduction. A higher external osmotic pressure is therefore not the explanation either. Both comparisons point to specific cation–carboxylate binding.

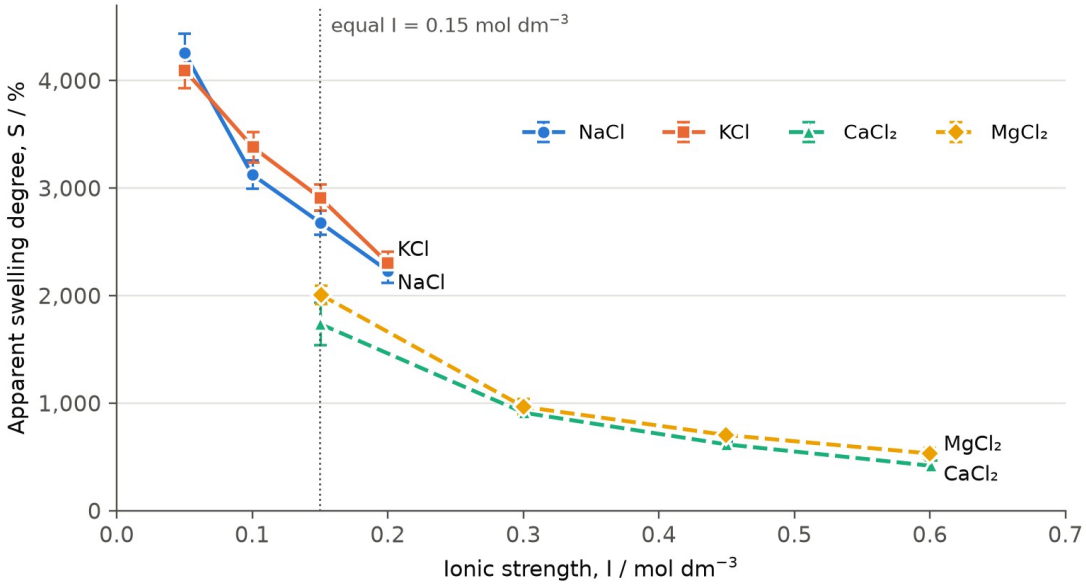


Figure 3. Mean apparent swelling degree against ionic strength for the non-zero concentrations. Error bars show the uncertainty in Table 9.

6.4 Differences within each charge group

$CaCl_2$ gave lower swelling than $MgCl_2$ at all four non-zero concentrations, by 13% on average (range 6–22%). The error bars separate at 0.15 and 0.20 mol dm^{-3} but overlap at 0.05 and 0.10 mol dm^{-3}. The difference is consistent with hypothesis 3: Ca^{2+} forms contact ion pairs with $–COO^-$, whereas the tightly bound hydration shell of Mg^{2+} (−1963 against −1616 kJ mol^{-1}) keeps it further from the carboxylate. Direct pairing produces stronger cross-links.

NaCl and KCl did not differ consistently: NaCl was higher at 0.05 mol dm^{-3} and lower at the other three concentrations, and the error bars overlap at three of the four. Stronger Na^+–carboxylate pairing may explain KCl being slightly higher at 0.10–0.20 mol dm^{-3}, but the average difference (4%) is similar to the measurement uncertainty, so these data cannot resolve it.

7. Conclusion

Increasing the concentration of every chloride salt from 0.00 to 0.20 mol dm^{-3} decreased the apparent swelling degree of sodium polyacrylate after 120 s. At 0.20 mol dm^{-3} it was 2230 ± 110% (NaCl), 2300 ± 100% (KCl), 533 ± 54% ($MgCl_2$) and 418 ± 45% ($CaCl_2$), compared with about 12,200% in distilled water.

The decrease followed a power law, $S = kc^n$. For the monovalent salts, n = −0.45 ± 0.07 (NaCl) and −0.39 ± 0.06 (KCl). These agree with Flory's theory in form but are 25–35% shallower than its limiting value of −0.60. This is partly because the fixed-charge concentration inside the gel (up to 0.25–0.48 mol dm^{-3}) was comparable to

the salt concentration, so the limiting condition was not reached, and partly because of the systematic errors in Section 8. The divalent salts gave much steeper exponents: n = −1.01 ± 0.16 ($CaCl_2$) and −0.95 ± 0.10 ($MgCl_2$).

Cation charge was the dominant factor. At 0.20 mol dm^{-3} the divalent salts gave 4 to 5.5 times lower swelling than the monovalent salts, and even at equal ionic strength (0.15 mol dm^{-3}) swelling was 31–35% lower. The effect of divalent cations is therefore more than extra screening, and is best explained by Ca^{2+} and Mg^{2+} bridging carboxylate groups. This agrees with Horkay, Tasaki and Basser (2000), who reported that small amounts of $CaCl_2$ make polyacrylate gels in NaCl solution shrink sharply, and with Mussel, Basser and Horkay (2019), who found that divalent cations lower the swelling pressure of these gels far more than monovalent cations. $CaCl_2$ gave on average 13% less swelling than $MgCl_2$, consistent with the stronger contact ion pairing of Ca^{2+}; NaCl and KCl could not be distinguished.

Hypotheses 1 and 2 are therefore supported, except for the monovalent gradient values, and hypothesis 3 only for Ca^{2+}/Mg^{2+}. In practice, a polyacrylate dressing would absorb far less of a fluid containing divalent ions than tests in water suggest.

8. Evaluation

8.1 Strengths

A freshly prepared solution in each of the 60 trials made every result independent, and the water controls agreed within 2.4% across all four series. The main findings are large compared with the uncertainties: at 0.20 mol dm^{-3} the monovalent and divalent means differ by a factor of four or more, while the uncertainties are 4–11%.

8.2 Weaknesses and improvements

The weaknesses are listed in order of their likely effect on the conclusion.

1. Solution retained by the tea bag (systematic). The empty bag was weighed dry, so any solution soaked up by the paper was counted as polymer swelling. This raises every S value by a fixed mass r, which affects low swelling values most. The data sets an upper limit: the smallest mass gain in any trial was 0.191 g ($CaCl_2$, 0.20 mol dm^{-3}), so $r < 0.19$ g. That is still up to 380 percentage points, as much as 90% of the $CaCl_2$ result at 0.20 mol dm^{-3}. Correcting for r would make all gradients steeper, but even the maximum possible correction only changes the NaCl gradient from −0.45 to −0.52, so bag retention alone cannot explain the difference from −0.60. Because the correction lowers the small divalent values proportionally most, it would widen, not remove, the difference between the charge groups. *Improvement:* treat three empty bags in each solution exactly as in steps 9–13 and subtract their mean mass gain, as recommended for the tea-bag method by Snoeck, Schröfl and Mechtcherine (2018).

2. 120 s may not reach equilibrium (systematic). If swelling was still increasing, every *S* value is below its equilibrium value. The gel has the most water still to absorb in the most dilute solutions, so the shortfall is probably largest there, which would also make the gradients shallower. This is the main threat to the charge conclusion: if divalent cations only *slowed* absorption, the gap could shrink at equilibrium. However, equilibrium studies (Horkay, Tasaki and Basser, 2000; Mussel, Basser and Horkay, 2019) show the same strong divalent effect, so a reversal is unlikely. *Improvement:* use a preliminary trial to measure *S* at 2, 5, 10, 20, 40 and 60 min in water and in 0.20 mol dm^{-3} $CaCl_2$ to find when mass stops changing (for example, less than 2% change between readings), then use that immersion time.

3. Removing surface liquid (random). "Until it stops dripping" depends on judgement, and blotting pressure was not controlled. This is the most likely cause of the highest relative standard deviations: 11.4% for $CaCl_2$ at 0.05 mol dm^{-3} and 10–11% for both divalent salts at 0.20 mol dm^{-3}, where the whole gain was only 0.19–0.29 g, so a few hundredths of a gram of surface liquid is significant. *Improvement:* hang each bag for a fixed 60 s, then blot once for a fixed time on the same grade of tissue without pressing.

4. Small polymer mass (random). The ±0.002 g uncertainty in *H* was about 98% of the propagated uncertainty. *Improvement:* in salt solutions, use 0.200 g of polymer, which reduces this contribution to 1%. In water, 0.050 g should be kept, because a larger sample would swell to over 20 g and would not fit freely in the bag.

5. Temperature not measured (systematic). Swelling depends on temperature, and "the same room" does not control it. The good agreement of the water controls suggests this effect was small. *Improvement:* carry out the immersion in a water bath at 25.0 ± 0.5 °C.

6. Equal molarity confounds charge with ionic strength. Only one point per salt could be compared at equal ionic strength (Section 6.3). *Improvement:* design a second experiment at matched ionic strengths, for example 0.15, 0.30 and 0.45 mol dm^{-3} NaCl against 0.05, 0.10 and 0.15 mol dm^{-3} $CaCl_2$.

8.3 Extensions

- Test body-fluid-like mixtures (0.15 mol dm^{-3} NaCl with 2.5×10^{-3} mol dm^{-3} $CaCl_2$) to see whether a little Ca^{2+} has a large effect.
- Test whether calcium cross-linking is reversible by transferring $CaCl_2$-treated gels into EDTA solution, which complexes Ca^{2+}.
- Use infrared spectroscopy to detect a shift in the carboxylate stretching absorption (about 1550–1610 cm^{-1}) when Ca^{2+} binds.

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